

Alvaro Perdomo Gutierrez

Machine Learning Engineer

alvaroperdomo121@outlook.com · +57 321 4105870 · Neiva, Huila, Colombia · www.linkedin.com/in/alvaro-perdomo-gutierrez-0466103b4/

PROFESSIONAL SUMMARY

Machine Learning Engineer with **12+** years across gaming, e-commerce, and healthcare, building production-grade ML systems for recommendation engines, demand forecasting, and marketplace optimization. Expertise in end-to-end workflows: data pipelines, model training with TensorFlow and PyTorch, deployment on AWS and cloud platforms, and real-time monitoring. Proven track record scaling ML platforms to millions of users while maintaining model performance and system reliability.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, Go, SQL, HTML, CSS

Generative AI: LangChain, LlamaIndex, OpenAI API, Hugging Face, Prompt Engineering

Data & ML: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, Spark, Airflow, MLflow, Weights & Biases, dbt

Backend: FastAPI, Django, Flask, Node.js, Express, Spring Boot, Kafka, RabbitMQ

Frontend: React, React Native, Next.js, Vue.js, Tailwind CSS, Redux

Database: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, DynamoDB, TcaplusDB, Cassandra

Testing: Pytest, Jest, Playwright, Selenium, k6

Cloud & DevOps: AWS, GCP, Azure, AWS Lambda, S3, CloudFront, EC2, RDS, SQS, CloudWatch, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins

Others: RESTful API, GraphQL, WebSocket, Grafana, Prometheus, Datadog, Git, GitHub, Jira, Confluence, Agile, Scrum

EXPERIENCE

Leanware

May 2024–Present

Senior Machine Learning Engineer

- Led the development of the AI fitness model for GloFlow using **Python**, **TensorFlow**, and **AWS** infrastructure, training and deploying computer vision models to analyze workout form and provide real-time corrective feedback to thousands of daily active users.
- Spearheaded the machine learning pipeline for the AI Grader, implementing **PyTorch**-based dental evaluation models with feature extraction and automated quality assessment that reduced manual review time by **40%** while maintaining clinical accuracy standards.
- Drove end-to-end development of the PRD Agent, leveraging **LangChain** and **OpenAI API** to build an AI-powered product requirements generator, integrating with GitHub via **Python** APIs and deploying on **AWS Lambda** to serve internal teams on-demand.

- Built data ingestion and preprocessing pipelines using **Pandas** and **NumPy** to normalize multi-source training data across fitness and medical domains, ensuring model robustness across diverse user populations and edge cases.
- Implemented **MLflow** and **Weights & Biases** experiment tracking for model versioning and hyperparameter tuning, streamlining the iteration cycle from weeks to days and enabling reproducible training runs across **AWS** EC2 clusters.
- Designed and deployed monitoring dashboards with **Prometheus** and **Grafana** to track model performance metrics, data drift, and prediction latency in production, enabling rapid detection of performance degradation.
- Collaborated with product and backend teams to define feature requirements and API contracts, ensuring deployed models integrated seamlessly with **FastAPI** and **React** frontends across multiple applications.
- Optimized inference performance by implementing batch prediction with **Redis** caching and model quantization, reducing per-prediction latency from **850ms** to **220ms** under peak traffic conditions.
- Evaluated and integrated third-party ML services for specialized tasks, selecting the optimal framework (**TensorFlow** vs. **PyTorch**) based on inference latency, model accuracy, and infrastructure constraints.
- Documented model architectures, training procedures, and deployment configurations in **Confluence**, enabling knowledge transfer across distributed teams and reducing onboarding time for new ML engineers by 25%.

ThredUp

Mar 2022–May 2024

Senior Software Engineer, Machine Learning

- Architected and deployed a visual recognition pipeline using **PyTorch** and **TensorFlow** to classify 100,000+ garments daily, training models on proprietary image datasets tagged by the Lily AI system that processed ~275K images weekly.
- Migrated the machine learning platform to **Databricks**, enabling distributed training and inference at scale, which reduced model deployment cycles from 2 quarters to 1 month and accelerated MVP development from 2 months to 2 weeks.
- Built end-to-end machine learning workflows for resale-pricing optimization using **Python**, **Pandas**, and **scikit-learn**, analyzing 172M+ items across 55K brands to predict optimal pricing and increase sell-through by **15%** and iOS conversion by **2.8%**.
- Implemented feature engineering pipelines with **Spark** and **SQL** to compute 500+ derived features from transaction history, user behavior, and garment metadata, feeding recommendation and ranking models serving 100K+ daily predictions.
- Designed and deployed the Image Search feature using embedding models trained with **TensorFlow**, leveraging **Elasticsearch** for fast similarity search and style-based retrieval that powered AI Shopping launched in August 2024.
- Developed automated data validation and model monitoring with **Prometheus** and custom **Python** scripts, detecting model drift and retraining triggers that prevented **8+** prediction quality issues from reaching production.

- Integrated **MLflow** for experiment management and model registry, establishing governance for model versioning and A/B testing framework that enabled teams to safely test new models against production baselines.
- Mentored junior engineers on machine learning best practices including cross-validation, hyperparameter tuning with **Optuna**, and statistical significance testing, improving model validation rigor across the platform.
- Collaborated with data science team to define success metrics and evaluate model improvements, balancing business objectives (sell-through, conversion) against computational constraints (**AWS** cost, inference latency).

Spring Health

Nov 2020–Mar 2022

Senior Software Engineer, ML Platform

- Led the scaling of a precision-matching machine learning platform that grew from supporting 2M+ members to 6x that volume, maintaining model accuracy and clinical validity while serving Fortune 500 clients including PepsiCo, Instacart, and Bain.
- Implemented operational machine learning models for dropout-risk prediction using **Python**, **scikit-learn**, and **Airflow**, identifying at-risk members with **78%** precision and enabling proactive interventions that reduced member churn by **12%**.
- Built demand forecasting pipelines using time-series models (**ARIMA**, **Prophet**) and **PyTorch** neural networks to predict utilization trends across 500+ enterprise accounts, informing infrastructure scaling and clinical staffing decisions.
- Engineered the Care Navigation platform for global expansion, integrating **PostgreSQL** and **Redis** to handle multi-jurisdiction matching rules and jurisdiction-aware care provider recommendations across 10+ countries by September 2021.
- Designed and deployed HIPAA-compliant data pipelines on **AWS** using **S3**, **RDS**, and **Lambda**, ensuring secure data handling for healthcare enterprise customers and passing compliance audits for 15+ Fortune 500 organizations.
- Developed feature engineering workflows with **Pandas** and **SQL** to construct 200+ features from clinical notes, member demographics, and provider availability, training ensemble models that improved matching accuracy by **18%**.
- Implemented continuous model monitoring using **Datadog** and custom dashboards tracking prediction accuracy, data quality, and system latency across production deployments, enabling rapid response to performance degradation.
- Collaborated with clinical teams to validate model outputs and establish decision rules ensuring healthcare compliance, balancing machine learning recommendations with clinical judgment and regulatory requirements.
- Mentored backend engineers on model integration patterns, establishing **REST API** standards for model serving and building inference SDKs that simplified integration with downstream systems.

Tencent Games

Nov 2014–Nov 2020

Machine Learning & Game Systems Engineer

- Built machine learning systems for Honor of Kings mobile MOBA, developing **Python**-based predictive models for player churn and engagement that enabled targeted retention campaigns for 100M+ daily active users by November 2020.
- Engineered data pipelines processing 40M+ daily events from **TcaplusDB** using **Spark** and **Python**, aggregating player behavior data to train matchmaking and difficulty-balancing models that improved new player retention by **22%**.
- Implemented anti-cheat machine learning models using kernel-level analysis and behavioral pattern detection, reducing fraudulent accounts by **35%** while maintaining false-positive rates below 0.5% for fair gameplay.
- Designed demand forecasting models using **scikit-learn** and **PyTorch** to predict server load and infrastructure requirements across regional servers, optimizing **AWS** and **GCP** resource allocation and reducing over-provisioning costs by **28%**.
- Built recommendation systems using collaborative filtering and content-based models trained on **Python** and deployed to serve personalized game recommendations and seasonal event suggestions to millions of players daily.
- Collaborated with game design teams to instrument telemetry and define success metrics, establishing data pipelines with **Kafka** that captured 10M+ daily events and enabled rapid iteration on game balance and progression systems.
- Contributed to platform infrastructure for multiple titles (PUBG Mobile, Arena of Valor, Contra Returns) by developing shared **Python** data utilities and model serving infrastructure on **GCP**, reducing model deployment overhead for new game launches.
- Mentored 5+ junior engineers on machine learning fundamentals, best practices for production ML systems, and handling massive scale with **Spark** and distributed computing frameworks.

EDUCATION

The Hong Kong Polytechnic University

Sep 2010–Nov 2014

Bachelor of Science in Computer Science